



**FACULDADE DE INFORMÁTICA E ADMINISTRAÇÃO
PAULISTA: ENGENHARIA DE DADOS**

Graph Databases and Analytics

SÃO PAULO
2026

29ABD

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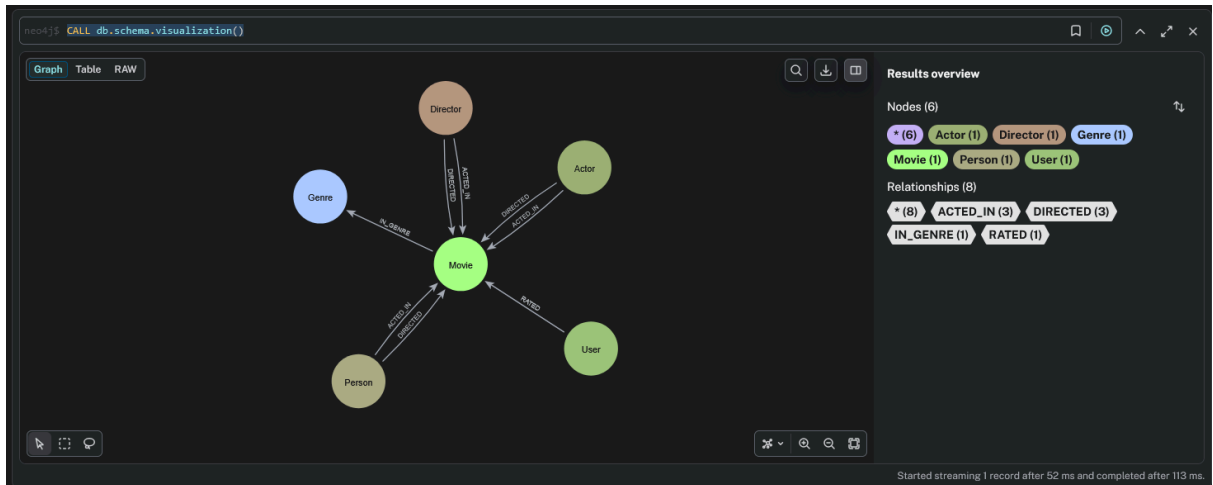
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Tutorial Movies Recommendations

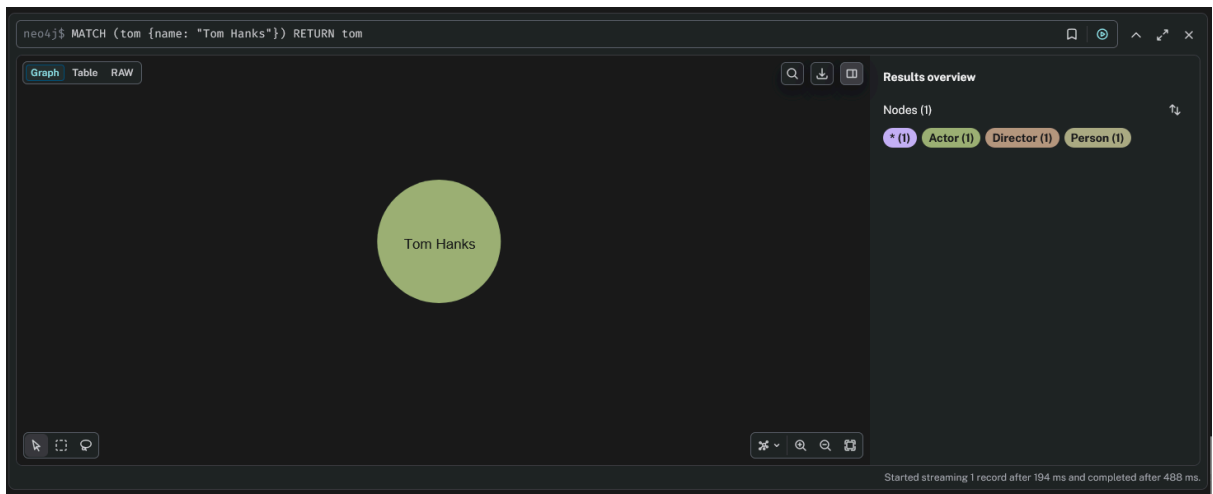
Ver o schema do grafo

CALL db.schema.visualization()



Buscar um ator por igualdade

MATCH (tom {name: "Tom Hanks"}) RETURN tom





Filmes de uma década com WHERE

MATCH (nineties:Movie) WHERE nineties.year >= 2010 AND nineties.year < 2020 RETURN nineties.title

The screenshot shows a Cypher query interface with a dark theme. The query is: `1 MATCH (nineties:Movie)
2 WHERE nineties.year >= 2010 AND nineties.year < 2020
3 RETURN nineties.title`. Below the query, there are tabs for 'Table' and 'RAW'. The 'Table' tab is selected, showing a list of movie titles. The titles are: "Water for Elephants", "African Cats", "Fast Five (Fast and the Furious 5, The)", "Something Borrowed", "Bridesmaids", "Priest", "Beaver, The", "Pirates of the Caribbean: On Stranger Tides", "Midnight in Paris", and "Kid With a Bike, The (Le gamin au vélo)". The interface also shows a status bar at the bottom: "Started streaming 1,362 records after 203 ms and completed after 293 ms."

nineties.title
"Water for Elephants"
"African Cats"
"Fast Five (Fast and the Furious 5, The)"
"Something Borrowed"
"Bridesmaids"
"Priest"
"Beaver, The"
"Pirates of the Caribbean: On Stranger Tides"
"Midnight in Paris"
"Kid With a Bike, The (Le gamin au vélo)"

Diretores de um filme via relacionamento

MATCH (cloudAtlas {title: "Valentine's Day"})<-[:DIRECTED]-(directors) RETURN directors.name

The screenshot shows a Cypher query interface with a dark theme. The query is: `1 MATCH (cloudAtlas {title: "Valentine's Day"})<-[:DIRECTED]-(directors)
2 RETURN directors.name`. Below the query, there are tabs for 'Table' and 'RAW'. The 'Table' tab is selected, showing a single record: "Garry Marshall". The interface also shows a status bar at the bottom: "Started streaming 1 record after 143 ms and completed after 368 ms."

directors.name
"Garry Marshall"

Co-atores do Tom Hanks

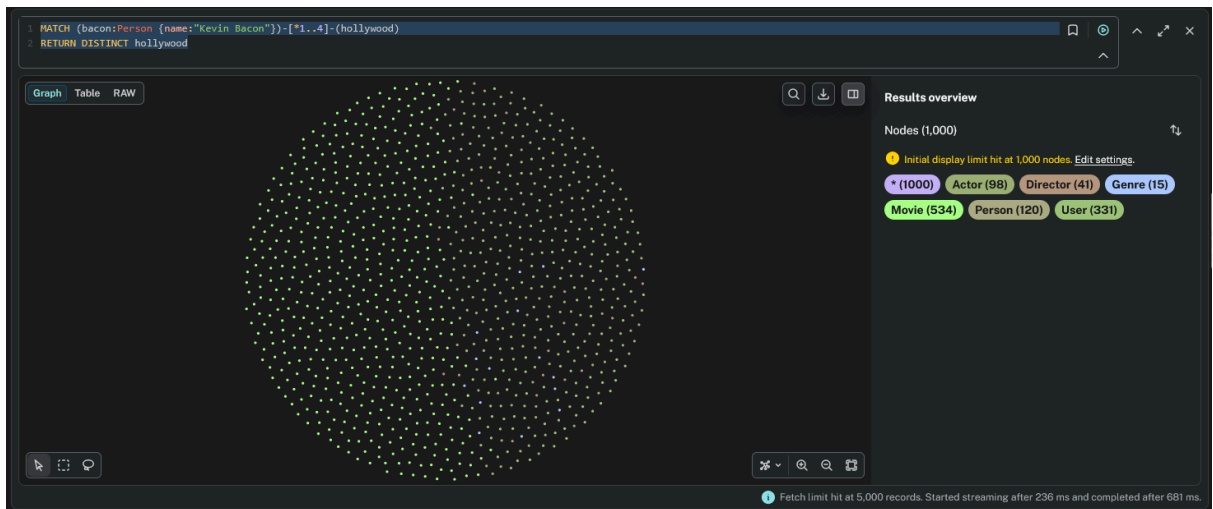
MATCH (tom:Person {name:"Tom Hanks"})-[:ACTED_IN]->(m)<-[:ACTED_IN]-(coActors) RETURN [coActors.name](#)

The screenshot shows a Cypher query interface with a dark theme. The query is: `1 MATCH (tom:Person {name:"Tom Hanks"})-[:ACTED_IN]->(m)<-[:ACTED_IN]-(coActors)
2 RETURN coActors.name`. Below the query, there are tabs for 'Table' and 'RAW'. The 'Table' tab is selected, showing a list of co-actors. The names are: "George Grizzard", "Tim Thomerson", "Rita Wilson", "John Candy", "Dabney Coleman", "Lori Singer", "Charles Durning", "Eugene Levy", "John Candy", and "Daryl Hannah". The interface also shows a status bar at the bottom: "Started streaming 114 records after 184 ms and completed after 188 ms."

coActors.name
"George Grizzard"
"Tim Thomerson"
"Rita Wilson"
"John Candy"
"Dabney Coleman"
"Lori Singer"
"Charles Durning"
"Eugene Levy"
"John Candy"
"Daryl Hannah"

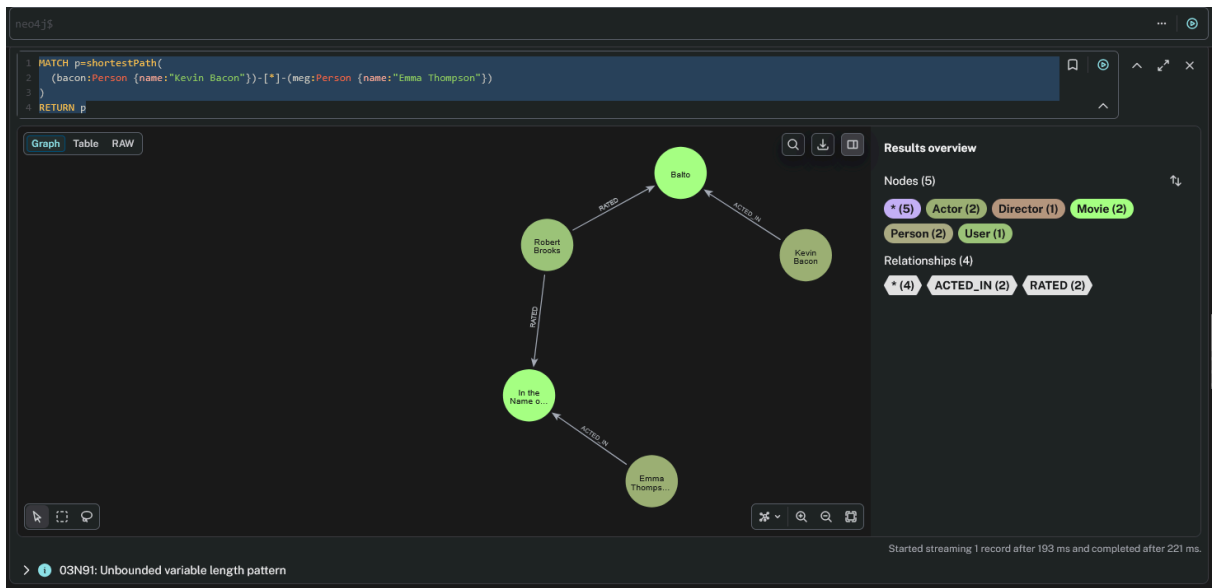
Exploração por distância (hops)

MATCH (bacon:Person {name:"Kevin Bacon"})-[*1..4]-(hollywood) RETURN DISTINCT hollywood



Menor caminho entre dois atores

MATCH p=shortestPath((bacon:Person {name:"Kevin Bacon"})-[*]-(meg:Person {name:"Emma Thompson"})) RETURN p



Avaliações de filmes Matrix

MATCH (m:Movie)-[:RATED]-(u:User) WHERE m.title CONTAINS 'Matrix' WITH m.title AS movie, COUNT(*) AS reviews RETURN movie, reviews ORDER BY reviews DESC LIMIT 5

movie	reviews
"Matrix, The"	259
"Matrix Reloaded, The"	82
"Matrix Revolutions, The"	54



Contagem de filmes por ano

MATCH (m:Movie) RETURN m.year as year, count(*) as total ORDER BY year DESC

The screenshot shows a query interface with a dark theme. The query is: `MATCH (m:Movie) RETURN m.year as year, count(*) as total ORDER BY year DESC`. The results are displayed in a table with two columns: 'year' and 'total'. The data is ordered by year in descending order. The interface includes a search bar, a 'Table' tab, and a 'RAW' tab. A status bar at the bottom indicates 'Started streaming 104 records after 96 ms and completed after 188 ms.'

year	total
1 null	29
2 2016	59
3 2015	180
4 2014	223
5 2013	221
6 2012	221
7 2011	229
8 2010	229
9 2009	254
10 2008	244

Agregações múltiplas

MATCH (m:Movie) RETURN m.year, count(*) as movie_count, min(m.imdbVotes) as min_votes, max(m.imdbVotes) as max_votes, sum(m.imdbVotes) as sum_of_votes, avg(m.imdbVotes) as average_votes

The screenshot shows a query interface with a dark theme. The query is: `MATCH (m:Movie) RETURN m.year, count(*) as movie_count, min(m.imdbVotes) as min_votes, max(m.imdbVotes) as max_votes, sum(m.imdbVotes) as sum_of_votes, avg(m.imdbVotes) as average_votes`. The results are displayed in a table with six columns: 'm.year', 'movie_count', 'min_votes', 'max_votes', 'sum_of_votes', and 'average_votes'. The data is ordered by year in descending order. The interface includes a search bar, a 'Table' tab, and a 'RAW' tab. A status bar at the bottom indicates 'Started streaming 104 records after 32 ms and completed after 98 ms.'

m.year	movie_count	min_votes	max_votes	sum_of_votes	average_votes
1 1995	270	41	981458	10713031	39677.892592592616
2 1994	239	31	1626900	10916456	45675.54811715478
3 1996	266	43	452146	8050591	30265.379699248115
4 1976	52	472	484769	1720896	33094.153846153844
5 1992	147	480	636075	5547244	37736.35374149659
6 1967	55	103	206837	1075027	19545.945454545443
7 1993	197	91	828323	7190864	36501.84771573605
8 1964	45	311	330672	1106694	24593.200000000004
9 1977	58	474	875151	1954628	33700.48275862069
10 1965	43	450	149581	735147	17096.44186046511

Top 20 duplas de atores que mais trabalharam juntos

```
MATCH (p1:Person)-[:ACTED_IN]->(m:Movie), (p2:Person)-[:ACTED_IN]->(m:Movie)
WHERE p1.name <> p2.name RETURN p1.name as name1, p2.name as name2, count(*)
as total ORDER BY total desc, name1, name2 LIMIT 20
```

name1	name2	total
"NirpU Maira"	"DEOULHU Maira"	11
"Masako Nozawa"	"Mayumi Tanaka"	10
"Mayumi Tanaka"	"Masako Nozawa"	10
"Bruce Balden"	"Jacqueline Bassett"	8
"Bruce Balden"	"Symon Basterfield"	8
"Jacqueline Bassett"	"Bruce Balden"	8
"Jacqueline Bassett"	"Symon Basterfield"	8
"Myrna Loy"	"William Powell"	8
"Symon Basterfield"	"Bruce Balden"	8
"Symon Basterfield"	"Jacqueline Bassett"	8

Started streaming 20 records after 290 ms and completed after 1,778 ms.

Conexões entre Jennifer Lawrence e Daniel Radcliffe (até 6 saltos)

```
MATCH path=(m:Person {name: 'Jennifer Lawrence'}) -[:ACTED_IN*6]- (n:Person {name:
'Daniel Radcliffe'}) RETURN path
```





Lab 2 - Amigos

Script Amigos

```
neo4j> create (Joaquin:Pessoa {nome:'Joaquin', idade:'30',sexo:'M'}) create (Beto:Pessoa {nome:'Beto', idade:'28',sexo:'M'}) create (Ana:Pessoa {nome:'Ana', idade:'26',sexo:'F'}) create (Pedro:Pessoa {nome:'Pedro', idade:'28',sexo:'M'})

neo4j> create (Joaquin:Pessoa {nome:'Joaquin', idade:'30',sexo:'M'})
neo4j> MATCH (Joaquin:Pessoa),(Beto:Pessoa)
neo4j> MATCH (Joaquin:Pessoa),(rock:Gosto)
neo4j> MATCH (Ana:Pessoa),(rock:Gosto)
neo4j> MATCH (Pedro:Pessoa),(rock:Gosto)
neo4j> MATCH (Joao:Pessoa),(rock:Gosto)
neo4j> MATCH (Beatriz:Pessoa),(rock:Gosto)
neo4j> MATCH (Beatriz:Pessoa),(Politica:Gosto)
neo4j> MATCH (Joao:Pessoa),(Politica:Gosto)
neo4j> MATCH (Joaquin:Pessoa),(Futebol:Gosto)
neo4j> MATCH (Ana:Pessoa),(Futebol:Gosto)
```

Cria o relacionamento "Joao curte Futebol"

MATCH (Joao:Pessoa),(Futebol:Gosto)

WHERE Joao.nome = 'Joao' AND Futebol.nome = 'Futebol'

CREATE (Joao)-[r:Like]->(Futebol)

;

```
neo4j> MATCH (Joao:Pessoa),(Futebol:Gosto) WHERE Joao.nome = 'Joao' AND Futebol.nome = 'Futebol' CREATE (Joao)-[r:Like]->(Futebol) ;

Created 1 relationship
03N9Q:Cartesian product
The disconnected pattern (Joao:Pessoa),(Futebol:Gosto) builds a cartesian product. A cartesian product may produce a large amount of data and slow down query processing.
MATCH (Joao:Pessoa),(Futebol:Gosto)
WHERE Joao.nome = 'Joao' AND Futebol.nome = 'Futebol'
CREATE (Joao)-[r:Like]->(Futebol)
;
```

Sugere amigos para Joaquim com base em gostos em comum

MATCH ({ nome: 'Joaquin' })-[:Like]->(gosto)-[:Like]-(pessoas)

WHERE NOT ({ nome: 'Joaquin' })-[:Amigo_DE]->(pessoas)

RETURN pessoas.nome, count(gosto)

ORDER BY count(gosto) DESC

;

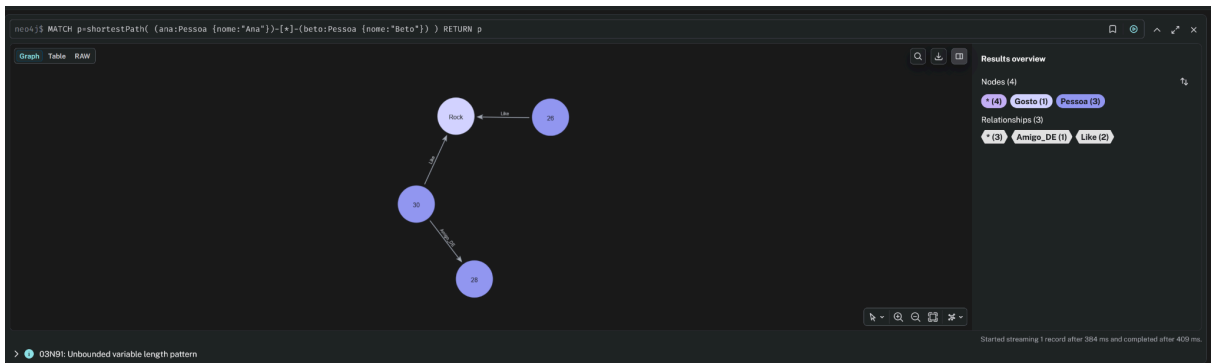
```
neo4j> MATCH ({ nome: 'Joaquin' })-[:Like]->(gosto)-[:Like]-(pessoas) WHERE NOT ({ nome: 'Joaquin' })-[:Amigo_DE]->(pessoas) RETURN pessoas.nome, count(gosto) ORDER BY count(gosto) DESC ;

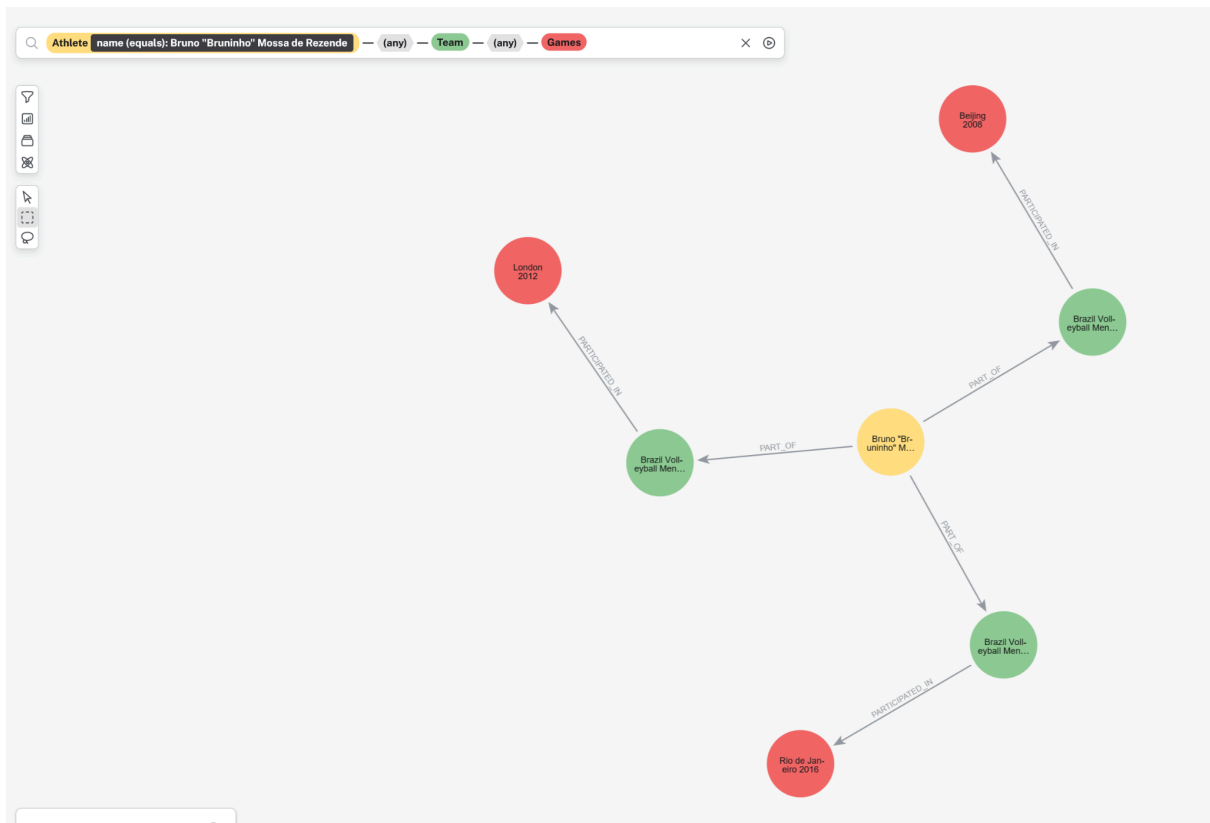
Table RAW
pessoas.nome      count(gosto)
"Ana"              2
"Joao"             2
"Pedro"            1
"Beatriz"          1

Started streaming 4 records after 1.987 ms and completed after 1.403 ms
```

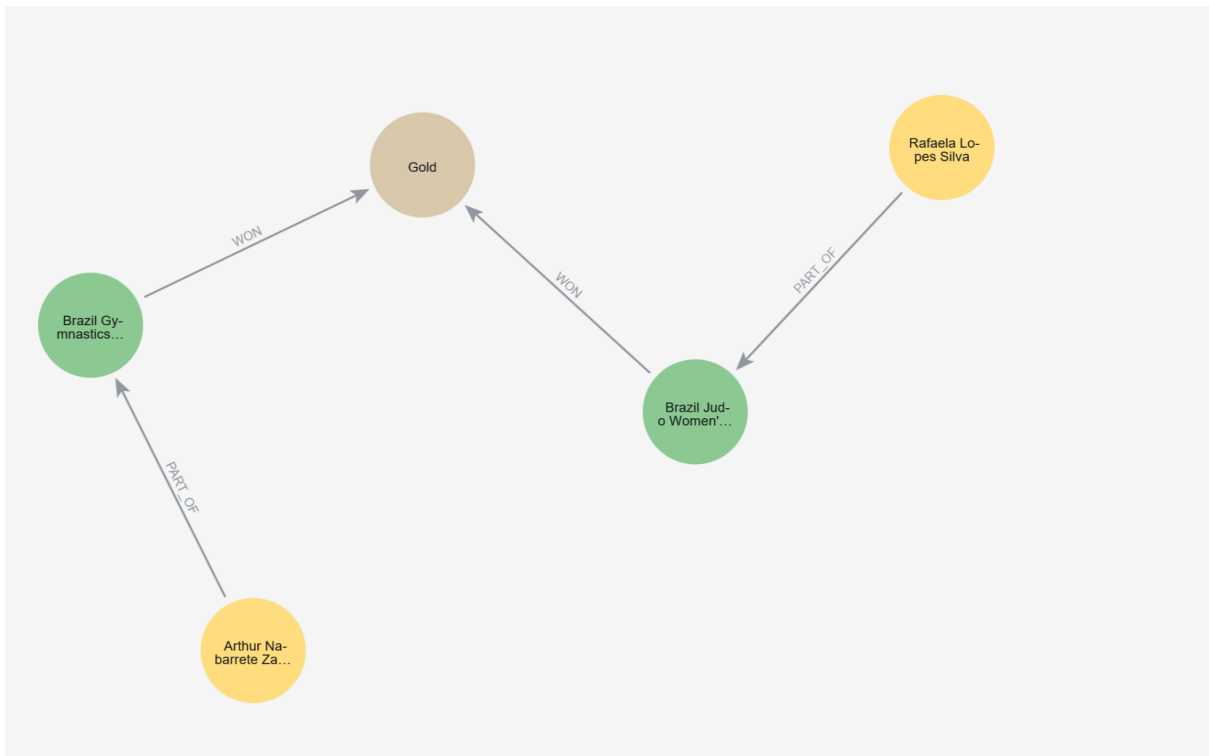

Encontra o menor caminho entre Ana e Beto no graf

```
MATCH p=shortestPath(
(ana:Pessoa {nome:"Ana"})-[*]-(beto:Pessoa {nome:"Beto"})
)
RETURN p
```

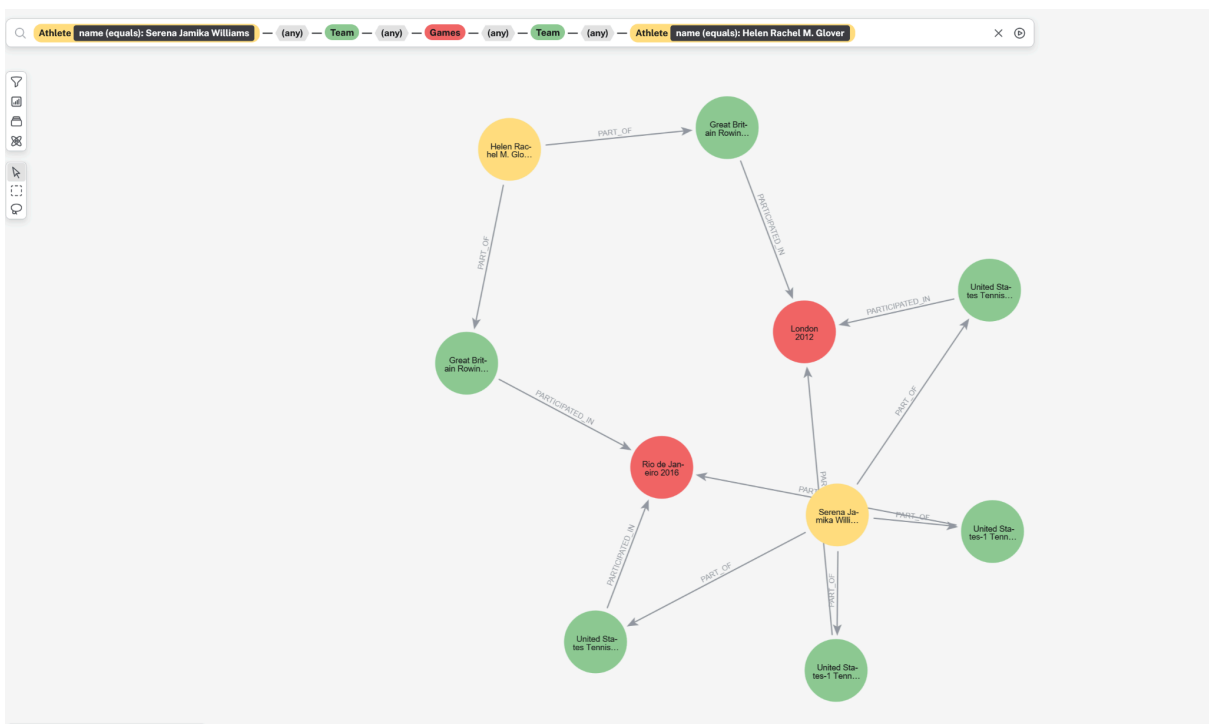




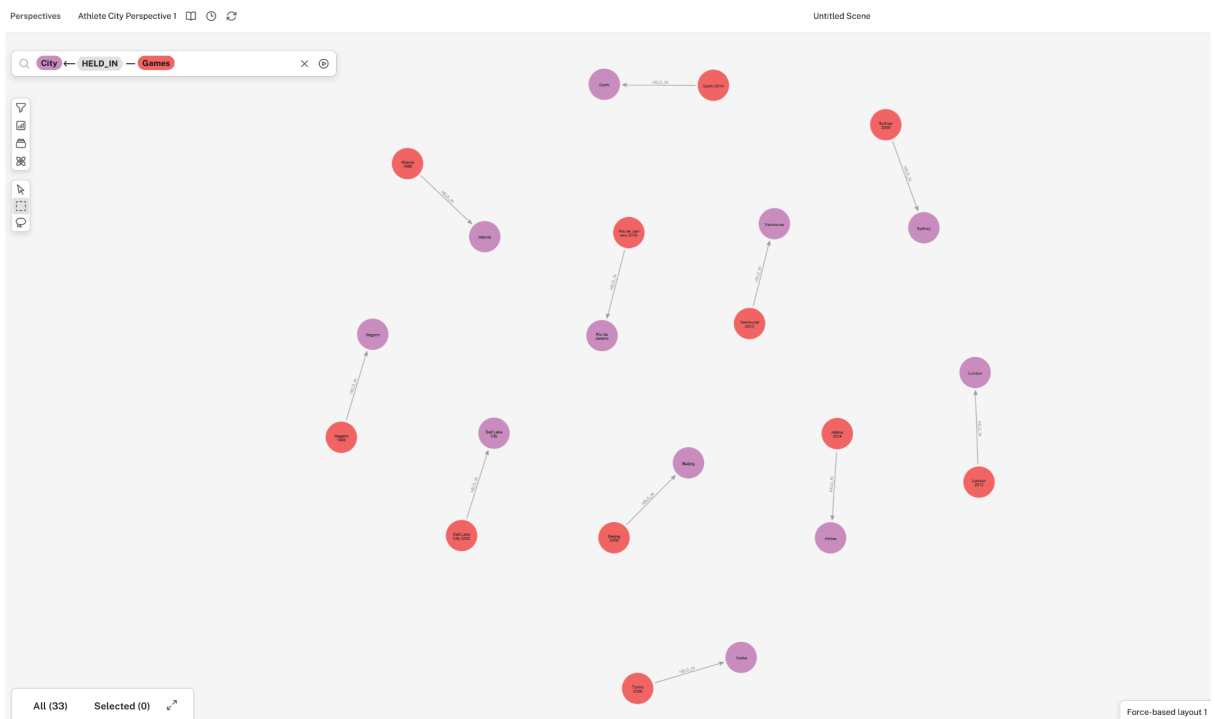
Qual o caminho mais curto entre Arthur Nabarrete Zanetti e a Rafaela Lopes Silva?



Quais jogos olímpicos unem as atletas Helen Glover e Serena Williams?



Quais cidades já sediaram os Jogos Olímpicos mais de uma vez?
Nenhuma



Quais atletas conquistaram mais de uma medalha de ouro?

